

Bootcamp Project: Polymer Chain Conformation under Different Solvent Conditions

The Liang Lab

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Goals

1. Perform molecular dynamics (MD) simulations of a coarse-grained polymer chain in a dilute solution using LAMMPS.
2. Analyze simulation data to quantify structural properties of polymer chains, such as radius of gyration and end-to-end distance.
3. Investigate how solvent quality influences polymer chain conformation.

Instructions

Step 0: Get Comfortable with the Linux Command Line

Before learning molecular dynamics (MD) simulations, you should first become comfortable using the Linux command line. Most MD simulations are run on high-performance computing (HPC) clusters, and these systems almost always use Linux. This means that even if you normally use Windows or macOS on your laptop, you will still need to know how to work in a Linux-style terminal in order to run simulations, organize files, submit jobs, and analyze results.

Do not worry if you have never used the command line before. This is very common, and you are not expected to know any programming at this stage. The goal of this step is simply to become familiar with basic terminal usage so that you can move between directories, create and edit files, and run simple commands with confidence.

If you use Windows Please install *Windows Subsystem for Linux (WSL)*. WSL allows you to run a Linux environment directly on your Windows computer, which is very helpful for practice and for future computational work.

If you use macOS No special installation is required. You can use the built-in *Terminal* application. Although macOS is not Linux, its command-line environment is similar enough for learning the basic skills you will need for HPC work.

What you should learn in this step By the end of this step, you should be able to do the following:

- open a terminal window and type commands,
- understand what a directory (folder) and file path are,
- move between directories using commands such as `pwd`, `ls`, and `cd`,
- create, copy, move, and delete files and directories,
- view the contents of a text file from the command line,
- edit a simple text file using a terminal editor, such as `nano` and `vim`,
- understand the idea of running a program by typing a command,
- connect to a remote machine using `ssh` (you do not need to master this yet, but you should know what it is).

Suggested tutorial To get started, please go through the following beginner-friendly tutorial:

<https://documentation.ubuntu.com/desktop/en/latest/tutorial/the-linux-command-line-for-beginners/>

As you work through the tutorial, do not just read it passively. Open your terminal and try each command yourself. The best way to learn the command line is by using it.

A note for beginners At this stage, it is completely normal to forget command names or make mistakes. Everyone does. You do not need to memorize everything immediately. What matters is that you begin to feel comfortable opening a terminal, trying commands, and understanding what they do.

Start the bootcamp To begin the bootcamp, please open a terminal window and download the bootcamp materials from GitHub using the command below:

```
git clone https://github.com/liang-softmatt/Bootcamp.git
```

This command creates a local copy of the bootcamp folder on your computer. After that, move into the downloaded folder by typing:

```
cd Bootcamp
```

Step 1: Learn LAMMPS

LAMMPS is an open source classical MD code. The best way to learn LAMMPS is to read its manual (<https://docs.lammps.org/Manual.html>), and practice by running simulations. In this bootcamp project, you will run simulations of a bead-spring polymer chain in an implicit solvent.

Install LAMMPS

- If you use Windows Subsystem for Linux (WSL): Open a terminal window, and run

```
sudo apt-get install lammps
```

This installs the LAMMPS executable named `lmp`. See https://docs.lammps.org/Install_linux.html#pre-built-ubuntu-and-debian-linux-executables for details.

- If you use macOS Terminal: Open a terminal window, and run

```
brew install lammps
```

This installs the executables `lammps_serial` and `lammps_mpi`. See https://docs.lammps.org/Install_mac.html for details.

Example LAMMPS input script Following is an example of LAMMPS input script, `in.run` (we often name the input script as `in.xxx`), for the simulation of this bootcamp.

```
1 # An example LAMMPS input script
2 variable      eps equal 0.1
3 variable      rcut equal 2.5
4 variable      T equal 1.0
5 variable      damping equal 10.0
6
7 units          lj
8 atom_style     molecular
9 dimension      3
10 boundary      p p p
11 neighbor       1.5 bin
12 neigh_modify   every 1 delay 0 check yes
13
14 read_data      data.chain
15 velocity       all create ${T} 4928459 rot yes dist gaussian
16
17 pair_style     lj/cut ${rcut}
18 pair_modify     shift yes
19 pair_coeff      1 1 ${eps} 1.0 ${rcut}
20
21 bond_style     fene
22 bond_coeff      1 30.0 1.5 1.0 1.0
23
24 special_bonds  fene
25
26 fix            1 all nve
27 fix            2 all langevin ${T} ${T} ${damping} 48279
28 timestep       0.01
29
30 compute        rg all gyration
31 fix            3 all ave/time 1000 1 1000 c_rg file gyration.dat mode scalar
32
33 dump            1 all custom 10000 dump.chain id mol type x y z ix iy iz
34 dump_modify     1 flush yes
35
36 thermo         1000
37 thermo_style    one
38 thermo_modify   flush yes
```

```

39
40 run          10000000
41
42 write_data    data.equil

```

Listing 1: An example of LAMMPS input script: `in.run`

Run LAMMPS It is always a good practice to run a simulation fresh in a new folder. Create a new folder, copy the input script (`in.run`) and input data file (`data.chain`) from the bootcamp folder.

- **If you use Windows Subsystem for Linux (WSL):** run the following command in your terminal

```
lmp -in in.run
```

- **If you use macOS Terminal:** run the following command in your terminal

```
lmp_serial -in in.run
```

What to expect after running LAMMPS If the simulation starts correctly, LAMMPS will print information to the terminal and create a log file, usually named `log.lammps`. According to our input script, it will also generate a text file recording the radius of gyration of the chain (`gyration.dat`, see line 30 and 31 in `in.run`), a LAMMPS dump file recording the trajectory of the chain (`dump.chain`, see line 33 in `in.run`), and a LAMMPS data file recording the final chain conformation (`data.equil`, see line 33 in `in.run`). If the run fails, read the error message carefully. In many cases, the error comes from a missing file, a typo in the input script, or running the command in the wrong folder.

Step 2: A Simple yet Powerful Model of Polymer: The Bead-Spring Model

The bead-spring model is the simplest model that captures the universal behavior of polymers. This model is introduced by Kurt Kremer and Gary Grest in their seminal paper *Dynamics of entangled linear polymer melts: A molecular-dynamics simulation*. (Kremer, K. and Grest, G.S., *J. Chem. Phys.* **1990**, 92(8), 5057-5086). In this model, many spherical beads (i.e., *monomers*) are connected by springs (i.e., *bonds*) to form a polymer chain. Non-bonded beads interact with each other through a truncated-shifted Lennard-Jones potential (defined in lines 17-19 in `in.run`). Bonded potential is described by the finite extensible nonlinear elastic (FENE) potential (defined in lines 21-22 in `in.run`).

Questions to Think About (Step 2)

Read LAMMPS manual and answer the following questions:

1. How does the Lennard-Jones potential change with different interaction strength (`{eps}` in `in.run`) and cutoff distance (`{rcut}` in `in.run`)? Why is it truncated and shifted in simulations?
2. Given the `bond_coeff` defined in `in.run`, what would be the equilibrium bond length?
3. What does `special_bonds fene` mean in `in.run` (at line 24)? Why is this setting important?

Step 3: A Basic MD Simulation

Understand the basics of a MD simulation.

- **Reduced units:** Instead of SI units, the coarse-grained model uses reduced units. The units of length, energy, and mass are σ , ϵ , and m , respectively. And the Boltzmann constant $k_B = 1$.
- **Periodic boundary condition (PBC):** The PBC is used by almost all MD simulations. By applying the PBC, we can model the behavior of bulk materials while simulating only a small representative region of the system. The PBC is set at line 10 in [in.run](#).
- **Explicit solvent vs implicit solvent:** In explicit solvent simulations, the solvent molecules are represented explicitly by individual solvent particles. In contrast, implicit solvent simulations do not include solvent particles; instead, the solvent's influence is incorporated through effective forces and thermostats.

In implicit solvent simulations, the effect of solvent is modeled by a Langevin thermostat (line 27 in [in.run](#)), where the damping parameter (`{damping}` in [in.run](#)) captures the viscosity of the solvent. The solvent quality is captured by the interaction strength between solute beads (i.e., monomers on the polymer chain).

- **Initialization:** The simulation box size, beads coordinates and bonded topology are initialized by reading an LAMMPS data file (line 14 in [in.run](#)). The velocities of beads are initialized at line 15 in [in.run](#). Read LAMMPS manual (https://docs.lammps.org/read_data.html) for the format of LAMMPS data file. Following is an example LAMMPS data file of a single chain with $N = 10$ monomers:

```
1 LAMMPS data file for a bead-spring chain with N=10 beads
2
3 10 atoms
4 1 atom types
5
6 9 bonds
7 1 bond types
8
9 -10.0 10.0 xlo xhi
10 -10.0 10.0 ylo yhi
11 -10.0 10.0 zlo zhi
12
13 Masses
14
15 1 1.0
16
17 Atoms
18
19 1 1 1 0.0 0.0 0.0
20 2 1 1 0.97 0.0 0.0
21 3 1 1 1.94 0.0 0.0
22 4 1 1 2.91 0.0 0.0
23 5 1 1 3.88 0.0 0.0
24 6 1 1 4.85 0.0 0.0
25 7 1 1 5.82 0.0 0.0
26 8 1 1 6.79 0.0 0.0
27 9 1 1 7.76 0.0 0.0
```

```

28 10 1 1 8.73 0.0 0.0
29
30 Bonds
31
32 1 1 1 2
33 2 1 2 3
34 3 1 3 4
35 4 1 4 5
36 5 1 5 6
37 6 1 6 7
38 7 1 7 8
39 8 1 8 9
40 9 1 9 10

```

Listing 2: An example of LAMMPS input data file: data.chain

The chain in the example data file is too short. Use python to generate a data file of a polymer chain with $N = 100$. Remember to change the box dimension (see Question 2)

- **Integrator:** LAMMPS uses velocity-Verlet integrator, which is a Symplectic integrator. The integrator is defined at line 26 in [in.run](#). The integration timestep is defined at line 28 in [in.run](#).

Questions to Think About (Step 3)

1. Why use reduced units in coarse-grained simulations? How to map back to real units?
2. To simulate a polymer chain in a dilute solution, what is the minimum dimension of the simulation box?

Step 4: Run a simulation

A standard simulation includes the following steps:

1. **Initialization:** Explained in Step 3.
2. **Relaxation:** The initial configuration is usually not in **thermodynamic equilibrium**, while most material properties are defined under equilibrium conditions. A relaxation run is therefore required to bring the system to equilibrium.
3. **Production:** After equilibration, a production run is carried out to collect sufficient simulation data for evaluating material properties. Simulation trajectory provides a lot useful information, and is saved by `dump` command in LAMMPS (lines 33-34 in [in.run](#)).

You've already practiced running simulations using LAMMPS in Step 1, but for a short chain with $N = 10$. Now it is time to run the simulation for a long chain with $N = 100$.

Step 5: Visualization and Analysis

Visualization

One advantage of computer simulations over experiments is that you can directly observe the motion of individual molecules. To do so, you can visualize the simulation trajectory (`dump.chain`) by OVITO (<https://www.ovito.org/>) or VMD (<https://www.ks.uiuc.edu/Research/vmd/>).

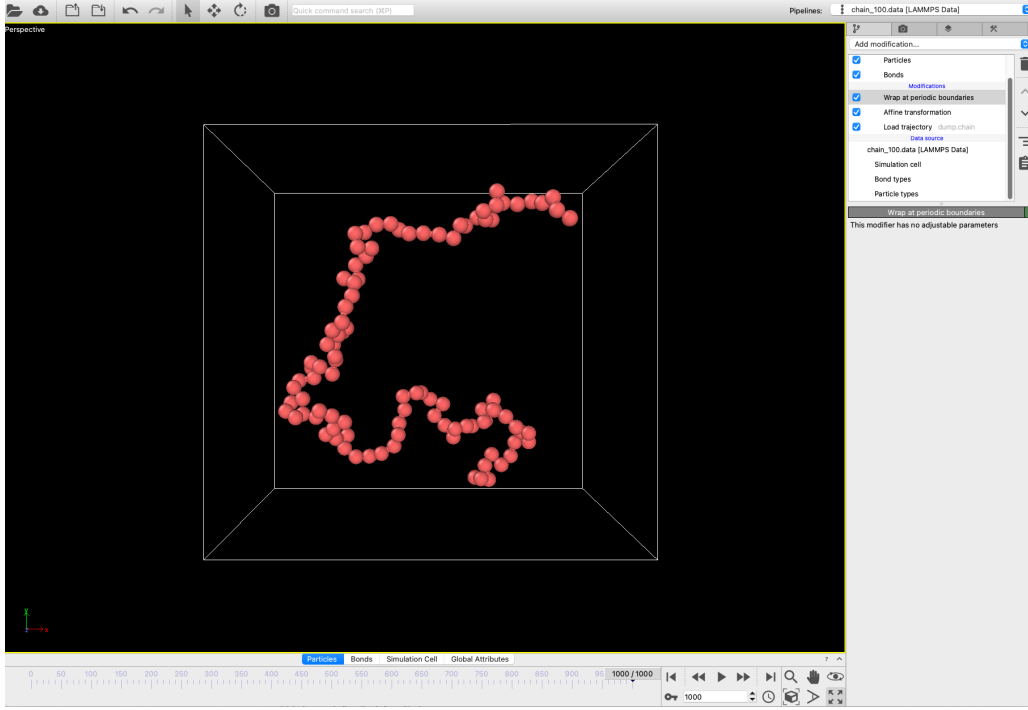


Figure 1: Screenshot of OVITO visualizing a chain with $N = 100$

Analysis

Analyze the chain conformation from the simulation trajectory of the production run.

Mean-squared radius of gyration:

$$\langle R_g^2 \rangle = \left\langle \frac{\sum_{i=1}^N m_i (\vec{r}_i - \vec{r}_{com})^2}{\sum_{i=1}^N m_i} \right\rangle \quad (1)$$

where $\vec{r}_{com}$ is the center of mass of the chain, $\vec{r}_i$ and m_i are the coordinate and mass of i -th bead, $\langle \dots \rangle$ represents a time average.

Mean-squared internal distance is the mean-squared distance between two monomers separated by n bonds on the same chain:

$$\langle R^2(n) \rangle = \langle (\vec{r}_{i+n} - \vec{r}_i)^2 \rangle \quad (2)$$

where $\langle \dots \rangle$ denotes both a time average and an average along the chain (e.g. $i = [1, N - n]$).

Questions to Think About (Steps 4 and 5)

1. How does LAMMPS record the simulation trajectory? What are **x**, **xs**, **xu**, **xsu** coordinates in LAMMPS dump file? What is **ix**? If a chain crosses the periodic boundary, how to treat its coordinates?

2. The initial conformation of the chain can be either a straight line (as in the example `data.chain.N5`) or a random coil. How does the initial conformation influence the simulation—for example, the number of steps required for relaxation or the equilibrium properties such as the radius of gyration? Use python to generate different initial conformations and test your hypothesis.
3. How long should a relaxation run be, and how can we determine whether the simulated system has reached equilibrium?
4. How to estimate error bars of $\langle R_g^2 \rangle$ and $\langle R^2(n) \rangle$? (Hint: block averaging)
5. How can solvent quality be inferred from chain conformation? (Hint: Plot $\langle R^2(n) \rangle$ vs n in log-log scale and fit the slope.)

Step 6: Run the Simulation under Different Conditions

Vary the pairwise interaction strength (change `$\epsilon_{ps}` in `in.run` from 0.1 to 1.0), and repeat Steps 4 and 5. Determine the conditions corresponding to good, theta, and poor solvents. Compare your results with the literature report (Graessley, W. W., Hayward, R. C., and Grest, G. S., *Macromolecules*, **1999**, 32, 10, 3510–3517).

Questions to Think About (Step 6)

1. Will $R^2(n)$ depends on the degree of polymerization of the chain, N ? If yes, why? Try to run simulations with different N under the same good solvent condition and verify your hypothesis.
2. Will the cutoff distance (`$\rcut$` in `in.run`) of the Lennard-Jones pairwise potential affect solvent quality? Will the tail correction affect chain conformation? Run simulations to test your hypothesis.

Step 7: Report the Results

Great work! You’ve made it to the end of this bootcamp!

You’ve learned how to use LAMMPS to simulate a polymer chain in implicit solvent and how to analyze its conformation from the resulting trajectory. The final step is to communicate your results effectively, just as you would at a scientific conference in front of colleagues and experts.

Below are guidelines to help you create clear and professional slides:

1. Start With an Introduction

- 1-2 slides of background information (i.e., chain statistics in different solvents)
- Clearly state the scientific question. What are you studying, and why does it matter?

2. Describe the Method Clearly

- Includes essential details of your simulation model so that others can reproduce it, such as
 - Force fields and parameters used
 - System size, box size

- Temperature and pressure control
- Integration algorithm, timestep
- Provide a schematic of your model. A clear simulation snapshot works great.

3. Summarize Your Analysis and Highlight Key Findings

- Show how you analyze the simulation trajectory. How are chain conformations characterized? ($\langle R^2(n) \rangle$, power-law fitting)
- Provide plots with readable labels and captions.
- Summarize your results. How does the chain conformation change with the interaction potential?
- *Convey insights, not just numbers.* Do your results agree with theoretical predictions, computational results from other references, and/or experimental observations? Why or why not? What is the molecular-level/microscopic mechanism?

4. Conclude With Take-Home Messages

- Summarize what you learned about polymer conformations in implicit solvent.
- Discuss limitations of the simplified model used in this project.
- Suggest possible next steps or follow-up simulations.

5. Tips

- Avoid long paragraphs; rely on bullet points.
- Use large figures with clear labels that are readable from a distance.
- Use figures/pictures when possible. A picture is worth a thousand words.
- Maintain consistent fonts, colors, and spacing throughout the presentation.
- For a 10-minute conference-style talk, aim for approximately 8-12 slides.